

SHIVARAJ SENTHIL RAJAN

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EDUCATION

Masters in Data Science - University of Colorado Boulder | 3.9/4.0

Aug '24 - May '26

Bachelors in Artificial Intelligence and Machine Learning - Anna University | 3.7/4.0

Aug '20 - May '24

Relevant Coursework: Big Data Architecture, Database Management Systems, Natural Language Processing, Statistical Learning, Data Mining, Information Visualization, Machine Learning, AI

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, PySpark, C++

Databases: PostgreSQL, MongoDB, Snowflake, BigQuery, Vector DBs (ChromaDB, Pinecone, FAISS)

Technologies/Frameworks: RAG, Agentic AI, LangChain, LangGraph, Tableau, Power BI, DAX, Cloud (AWS, GCP, Azure), Docker, Kubernetes, MLflow, FastAPI, CI/CD, Git, Agile

Data Engineering & Libraries: Spark, Kafka, Airflow, Databricks, dbt; scikit-learn, XGBoost, PyTorch, TensorFlow, Transformers, Pandas, NumPy

Analytical & BI Skills: Predictive Modeling, Time Series, A/B Testing, Dimensional Modeling, KPI Reporting, Feature Engineering, EDA, Stakeholder Communication

PROFESSIONAL EXPERIENCE

Data Analyst ,University of Colorado Boulder | *Boulder, CO*

Aug '24 - May '26

[University of Colorado Boulder Conference Services supports campus event operations, venue coordination, resource planning, and service delivery for academic, administrative, and community programs]

- Reduced procurement over-ordering by **15%** by analyzing **100+ SKU** consumption trends, translating seasonal demand patterns into clearer purchasing recommendations for non-technical department coordinators.
- Reconciled vendor pricing discrepancies by validating unit-cost differences, invoices, purchase benchmarks, and budget records, improving accuracy before quarterly financial review cycles.

AI/ML Engineering Intern, PM Accelerator | *Miami, FL*

May '25 - Nov '25

[PM Accelerator builds AI-powered product career platforms that support product teams, learners, and professional communities through technology-enabled learning and engagement]

- Improved document search and insight retrieval by **40%** by building **RAG pipelines**, **LLM workflows**, and **vector search** solutions for faster knowledge discovery across large document collections.
- Developed **NLP classification models** with **95% accuracy** by applying **Python**, machine learning, and behavioral feature analysis to improve personalization and recommendation quality.
- Modeled churn risk with **92% accuracy** using **supervised learning**, **feature engineering**, and validation techniques, helping product teams identify at-risk users and support retention strategies.

Data & ML Engineer, Sandron Impex Private Limited | *Bengaluru, India*

Feb '23 - May '24

[Sandron Impex Private Limited supports logistics, trading, procurement, vendor coordination, and operational decision-making across business workflows in India]

- Enhanced invoice risk detection by engineering **anomaly detection** models with calibrated risk thresholds, identifying **92%** of suspicious transactions and helping finance teams prioritize high-risk reviews.
- Automated reporting data quality improvements by building **ETL validation** checks, exception handling, and structured outputs, increasing reporting reliability by **25%** for monthly business reviews.
- Forecasted procurement demand using **ARIMA** and **Prophet** time-series models on historical operational data, reducing demand-planning cycle time by **15%** and supporting faster inventory replenishment decisions.

PROJECTS

AI Observability Platform on GKE [\[Link\]](#) | *Airflow, ChromaDB, Gemini, RAG, sentence-transformers*

- Helped data science teams diagnose **ML pipeline failures** and model issues faster by building an **AI observability assistant** on GKE that reads **Airflow** logs, source code, and pipeline metadata to deliver instant root-cause explanations.
- Engineered a semantic **RAG** retrieval engine over **4 ChromaDB collections** (350+ docs) using **sentence-transformers** (all-MiniLM-L6-v2, 384-dim), injecting live **Airflow task state** and **Kubernetes pod status** as real-time context before each **Gemini 2.5 Flash** call.

DataPilot AI Pro - Autonomous Multi-Agent Data Science Platform [\[Link\]](#) | *LangGraph, PPO RL, LLaMA 3.1, SHAP*

- Eliminated manual data science effort by building an autonomous **multi-agent ML platform** that converts any raw CSV upload into cleaned data, trained prediction models, **SHAP** explanations, and BI dashboards without any technical expertise required.
- Designed a **7-agent LangGraph** state-machine using **LangChain** and **LLaMA 3.1** (8B via **Ollama**), paired with a **PPO** agent (**Stable-Baselines3**, **Gymnasium**) trained on **30+ OpenML datasets** using **40 meta-features**, producing voting ensembles with **85%+** accuracy.

TrustLens AI - Real-Time Toxicity Detection [\[Link\]](#) | *BERT, FastAPI, Chrome Extension, Transformers*

- Improved online content safety for moderation teams by building a real-time **toxicity detection** system that classifies harmful Reddit comments across **6 categories** directly in the browser at **2.73+ comments per second** and **45ms** latency.
- Fine-tuned a **BERT** multi-label classifier (**unitary/toxic-bert**) with **weighted cross-entropy loss** and **GPU inference**, deployed via async **FastAPI** and **Manifest V3 Chrome Extension** for real-time toxicity scoring.

RESEARCH & PUBLICATIONS

Smart Fields: Enhancing Agriculture with Machine Learning [\[Link\]](#) - *IEEE AIMLA 2024*